



Unique phase change behaviors of ammonia droplets under varying ambient water vapor concentrations and pressures: A molecular dynamics simulation study

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ARTICLE INFO

Keywords:

Ammonia potential model
Phase change
Ambient gas composition
Molecular dynamics simulation

ABSTRACT

This study employs molecular dynamics (MD) simulations to study the effect of polar water vapor on the phase change characteristics of ammonia droplets under varying ambient pressures. First, a new flexible potential model for ammonia was developed based on first-principles calculations. Then, the accuracy of this model in predicting the thermodynamic and transport properties of ammonia was extensively validated. Consequently, MD simulations using the new potential model were conducted to explore phase change behaviors of ammonia droplets under different ambient environments. The results reveal that both elevated ambient pressures and increased water vapor concentrations can promote the ammonia droplet evaporation. A unique phase change behavior of ammonia droplets in nitrogen/water environments was observed. Specifically, the polar water vapor dissolves and subsequently condenses within the ammonia droplet, thereby facilitating a transition from the ammonia-dominated evaporation to the water-dominated evaporation. Moreover, the dissolution and condensation become more intense at higher initial water vapor concentrations or pressures. Finally, the specific mechanisms by which water vapor enhances ammonia droplet evaporation were explored. During the dissolution and condensation process, water vapor releases latent heat and increases thermal conductivity, raising the droplet temperature. Additionally, water weakens the hydrogen bonding among ammonia molecules, thereby lowering the evaporation energy barrier. These findings provide essential insights into the phase change mechanisms of liquid ammonia and their dependence on ambient conditions.

1. Introduction

With the shift toward energy decarbonization, ammonia, as a zero-carbon renewable fuel, is currently receiving considerable attention in the combustion community. It offers several practical advantages for engine applications, including high volumetric energy density, low storage and transportation costs, favorable safety, mature production technologies, and established transport infrastructures. Among injection methods, direct injection of liquid ammonia has been demonstrated to provide more efficient fuel delivery [1]. However, experimental measurements revealed that the phase change behavior of ammonia spray is highly sensitive to ambient conditions [2]. This inspires numerical investigations for assessing and improving ammonia spray vaporization models [3,4]. Unfortunately, a deeper understanding of ammonia phase change mechanisms is lacking, especially under different gas composition environments.

Previous studies have shown that several properties of ambient gases, such as molecular weight, molecular polarity, thermal conductivity, critical properties, and binary diffusion coefficient [5–7],

significantly affect liquid phase change processes. However, these studies were limited to some specific conditions, leaving it unclear whether the same principles apply to ammonia under engine-relevant subcritical conditions. Meanwhile, the potential complex coupling effects between ambient gas composition and pressure remain to be revealed. Moreover, in diesel-ammonia dual-fuel engine applications, a late injection strategy for liquid ammonia is typically employed to enhance combustion stability [8], resulting in the injection of ammonia concurrent with ongoing diesel combustion. Given that water vapor constitutes a major combustion product, its influence on the phase change behavior of ammonia needs further investigation. This is critical for advancing the fundamental understanding of the complex phase change mechanisms for ammonia under engine-relevant conditions.

Compared to traditional methods, molecular dynamics (MD) offers a powerful approach to explore complex phase change behaviors by capturing nanoscale details and underlying mechanisms with fewer assumptions and empirical approximations [9]. Chakraborty et al. [7]

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employed MD to study the supercritical evaporation of *n*-heptane in environments containing water or helium, finding that higher water concentrations lower the mixture critical pressure and promote supercritical transition, whereas helium has little effect. Chen et al. [5] investigated the influence of hydrogen-rich environments on ammonia phase change behaviors, revealing that hydrogen suppresses the cooling effect of ammonia, enhancing evaporation and advancing the transition of mixing mode toward diffusion. Recently, Zhang et al. [6] explored the supercritical transition characteristics of ammonia droplets in multi-component gas environments, pointing out that water vapor, hydrogen, and carbon dioxide can accelerate the supercritical transition of ammonia. However, given the high critical pressure of ammonia, the specific effects of water vapor on the phase change characteristics of ammonia droplets under conventional subcritical conditions and their influence mechanisms have not been comprehensively evaluated.

The accuracy of MD simulations relies heavily on the chosen potential model. However, the currently widely adopted rigid model [10], which fixes molecular bond lengths and bond angles, overlooks the significant role of intramolecular flexibility in determining liquid structure, dynamics, and thermodynamics [11]. Furthermore, imposing molecular rigidity via the SHAKE algorithm increases the computational cost and necessitates additional corrections when evaluating properties related to molecular degrees of freedom. These limitations highlight the need for a flexible potential model that can improve accuracy in simulating ammonia phase change behaviors while also reducing computational complexity.

To address these challenges, this study first develops a new flexible potential model for ammonia molecules based on first-principles calculations. This model is then employed in MD simulations to systematically investigate the phase change characteristics of ammonia droplets under varying water vapor concentrations and pressures.

2. MD simulation

2.1. New NH_3 flexible potential model

Potential models are typically developed either by fitting to experimental data or by deriving parameters from first-principles calculations. In this study, a new flexible potential model for ammonia was constructed based on first-principles calculations. Due to space limitations, detailed information on the model development can be found in the Supplementary Material.

The present model adopts four interaction sites to balance the model complexity and computational efficiency. All interaction sites coincide with the positions of the atomic nuclei, with the N atom assigned a negative partial charge and the three H atoms carrying positive charges. The intramolecular bond stretching potential U_b and the angle bending potential U_θ are represented by the Morse potential and the harmonic potential, respectively:

$$U_b = D_e (1 - \exp(-\beta(r_i - r_i^{\text{eq}})))^2 \quad (1)$$

$$U_\theta = k_\theta (\theta_i - \theta_i^{\text{eq}})^2 \quad (2)$$

where D_e represents the bond stretching potential well depth; β is the potential well flatness parameter, defined as $\beta = \sqrt{k_b/2D_e}$; k_b and k_θ denote the force constants for bond stretching and angle bending, respectively; r_i^{eq} and θ_i^{eq} are the equilibrium bond length and bond angle, respectively. Notably, the Morse potential is adopted for N-H bond stretching instead of the more common harmonic potential. Unlike the harmonic form, which accurately describes vibrations only near the equilibrium point, the Morse potential captures the energy variation across the entire bond length range, thereby providing a more realistic representation of bond stretching.

Table 1

New ammonia flexible potential model parameters.

Potential	Parameter		
Non-bond	ϵ (kcal/mol)	σ (Å)	q (e)
N	0.3568	3.385	-1.0194
H	—	—	0.3398
Bond	D_e (kcal/mol)	β (Å ⁻¹)	r_i^{eq} (Å)
N-H	118	2	1.015
Bond angle	k_θ (kcal/mol rad ²)	θ_i^{eq} (°)	
H-N-H	39.47	107.5	

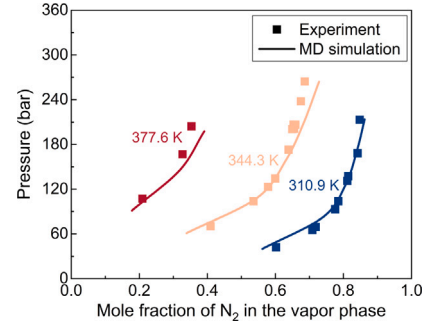


Fig. 1. Comparison of MD simulation and experimental multi-component VLE data [12].

For the non-bonded intermolecular potential $U_{\text{non-bond}}$, both the Lennard-Jones (LJ) 12-6 potential and Coulomb interaction are employed:

$$U_{\text{non-bond}} = 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} \quad (3)$$

where ϵ_{ij} and σ_{ij} are the LJ energy and size parameters, respectively; r_{ij} is the distance between atoms i and j ; q_i and q_j are the partial atom charges; ϵ_0 denotes the dielectric constant. In addition, the LJ parameters between different atoms are derived by the Lorentz-Berthelot mixing rules:

$$\epsilon_{ij} = \sqrt{\epsilon_i \epsilon_j}, \quad \sigma_{ij} = \frac{1}{2} (\sigma_i + \sigma_j) \quad (4)$$

The complete set of parameters for the flexible potential model is summarized in Table 1.

The thermodynamics vapor-liquid equilibrium (VLE) properties, including saturated liquid and vapor density, saturated vapor pressure, enthalpy of vaporization, and surface tension, and transport properties such as the self-diffusion coefficient in both liquid and gas phases, were systematically validated over a wide temperature range. The specific validation methods and results can be found in the Supplementary Material. In addition, the model reliability in predicting the multi-component VLE properties of the NH_3 - N_2 system was further verified. As shown in Fig. 1, the simulation results exhibit good agreement with experimental VLE data reported by Reamer et al. [12], thereby validating the MD setup and supporting its application in ammonia droplet evaporation simulations.

2.2. Simulation settings

This study systematically investigates the phase change characteristics of ammonia droplets in different gas environments. Detailed information on potential models for various ambient gases is summarized in the Supplementary Material. Initially, the droplet was equilibrated at 303 K using the canonical (NVT) ensemble. Similarly, the surrounding gas environments – pure nitrogen or nitrogen/water mixtures – were equilibrated under the NVT ensemble at their respective target temperatures and pressures, where different pressures were achieved by adjusting the number of gas molecules.

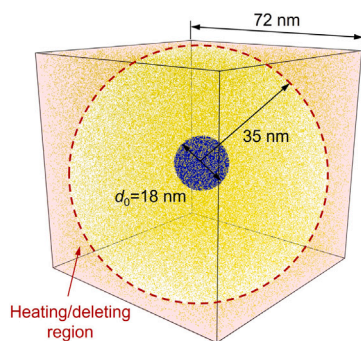


Fig. 2. Initial configuration for droplet evaporation (ammonia molecules are represented in blue, while nitrogen molecules are represented in yellow).

Following equilibration, the droplet was positioned at the center of the gas box, and any overlapping molecules were removed to generate the initial configuration for droplet evaporation, as illustrated in Fig. 2. Subsequent evaporation simulations were conducted using the microcanonical (NVE) ensemble in a cubic simulation box with a side length of 72 nm, allowing spontaneous heat and mass transfer. A spherical region with a radius of 35 nm was defined as the evaporation domain, while the outer region served as the heating region. Based on sensitivity analyses of droplet size (see Supplementary Material for details), an initial droplet diameter (d_0) of 18 nm was selected to balance the computational accuracy and efficiency. To maintain the target gas temperature, a Langevin thermostat was applied in the heating region. Additionally, evaporated ammonia molecules entering this region were removed to prevent vapor accumulation, elevated local pressure, and recondensation. A cutoff radius of 12 Å was adopted to achieve a compromise between accuracy and efficiency. Each case was run with a time step of 1 fs over a total duration of 30 ns, ensuring complete capture of the dynamic droplet evaporation process. All MD simulations were performed using the Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS).

Given the critical properties of ammonia (critical temperature of 405.6 K and critical pressure of 113.6 bar), an ambient temperature (T_a) of 1000 K was selected, with ambient pressures (P_a) varying from 50 to 200 bar. According to similar studies [6,7], nitrogen-based environments containing different mole fractions of water (0%, 10%, and 25%) were considered. For brevity, detailed procedures for calculating statistical droplet properties can be found in the Supplementary Material.

3. Results and discussion

3.1. Evaporation processes under different ambient conditions

Figs. 3(a) and 4(a) provide evaporation snapshots and evolutions of the squared normalized droplet diameter (d/d_0)² for ammonia droplets under varying ambient water vapor concentrations and pressures. Fig. 5 shows the quantitative results of evaporation rate (slope) and droplet lifetime (intercept) obtained by further fitting these (d/d_0)² curves. As expected, increasing ambient pressure accelerates the evaporation rate and shortens the droplet lifetime. This behavior is primarily driven by the combined effects of reduced latent heat of vaporization, a significant decrease in surface tension, and increased solubility of the surrounding gas in the liquid phase [9]. However, it can be observed that the addition of water vapor can also enhance ammonia droplet evaporation, with the enhancement becoming more pronounced at higher initial water concentrations and pressures. Moreover, no initial heating period is observed during the early stage of evaporation. This can be explained by the rapid evaporation of ammonia, which produces a volumetric evaporation rate that exceeds the volumetric expansion

rate. This phenomenon is consistent with the single-droplet evaporation experiments of Khuong et al. [13], who reported it as a characteristic feature of low-boiling-point fuels. These findings suggest a complex and distinctive phase change mechanism for ammonia droplets governed by the interplay of ambient water vapor and pressure.

A notable observation is that the (d/d_0)² curves in nitrogen/water environments clearly deviate from the classical d^2 law. Instead of showing a linear decline during the quasi-steady stage, these curves exhibit a distinct transition point in the latter half of the process (refer to Fig. 4(a)). After this point, the evaporation rate decreases noticeably, even leading to a longer droplet lifetime, as shown in Fig. 5. This trend aligns with the experimental observations of Khuong et al. [13] and the numerical findings of Vesala et al. [14], both of which demonstrated that ambient water vapor can significantly influence the evaporation process of ammonia droplets. In contrast, such behavior is not observed in pure nitrogen environments.

To further investigate the underlying mechanisms, a series of *n*-heptane droplet evaporation simulations were conducted under the same conditions (i.e., initial droplet diameter, initial droplet temperature, and ambient conditions) for comparison, as presented in Figs. 3(b) and 4(b). As can be seen, for ammonia droplets, a substantial number of water molecules adsorb on the droplet surface and subsequently dissolve into the droplet interior. These molecules then condense, eventually forming a water nucleus. In contrast, virtually no water molecules are observed to dissolve and condense within *n*-heptane droplets. These distinct phase change behaviors stem from fundamental differences in the molecular polarity. Specifically, substances with similar polarities tend to be more miscible. Similar condensation and dissolution phenomena of water vapor have also been reported for high-polarity short-chain alcohol droplets evaporating in humid air [15]. This inference is further corroborated by the quantitative analysis of the water mole fraction (x_{H_2O}) within the different droplets in nitrogen/water environments, as shown in Fig. 6. It can be observed that the water mole fraction in ammonia droplets increases significantly over time, while remaining nearly constant in *n*-heptane droplets. Due to the near insolubility of polar water vapor in non-polar *n*-heptane droplets, the (d/d_0)² curves for environments with different water vapor concentrations almost completely overlap, as depicted in Fig. 4(b). This reveals that the presence of water vapor in the gas environment has a negligible effect on *n*-heptane droplet evaporation. Consistently, the experimental results of Law et al. [15] also confirmed that humid air exerts no influence on the evaporation of low-polarity alcohols or non-polar alkanes.

3.2. Dissolution and condensation effects of water vapor

The preceding analysis highlights the critical influence of water-rich environments on the phase change behavior of ammonia droplets. To more intuitively illustrate this process, Fig. 7 provides the evolution of the mole fraction for each component within the droplet in the nitrogen/water environment, alongside the corresponding (d/d_0)² curve. Initially, the mole fraction of ammonia (x_{NH_3}) decreases steadily, while the mole fraction of water (x_{H_2O}) increases rapidly. It is evident that water vapor continuously dissolves into the droplet from the ambient gas, gradually transforming the initially pure ammonia droplet into an ammonia–water mixture. As evaporation proceeds, ammonia, with a higher volatility, evaporates preferentially, leading to a further decline in x_{NH_3} until it eventually reaches zero. Simultaneously, x_{H_2O} increases to unity, indicating that the residual water condenses into nuclei. That is, the mixture droplet is further transformed into a small water droplet and then continues to evaporate (see Fig. 3(a)). Consequently, the extended droplet lifetime in the nitrogen/water environment observed in Fig. 5(b) is primarily attributable to the residual water droplet, rather than the original ammonia content. In other words, although the total droplet lifetime is prolonged in the nitrogen/water environment, the ammonia component evaporates more rapidly, as shown in Fig. 3(a).

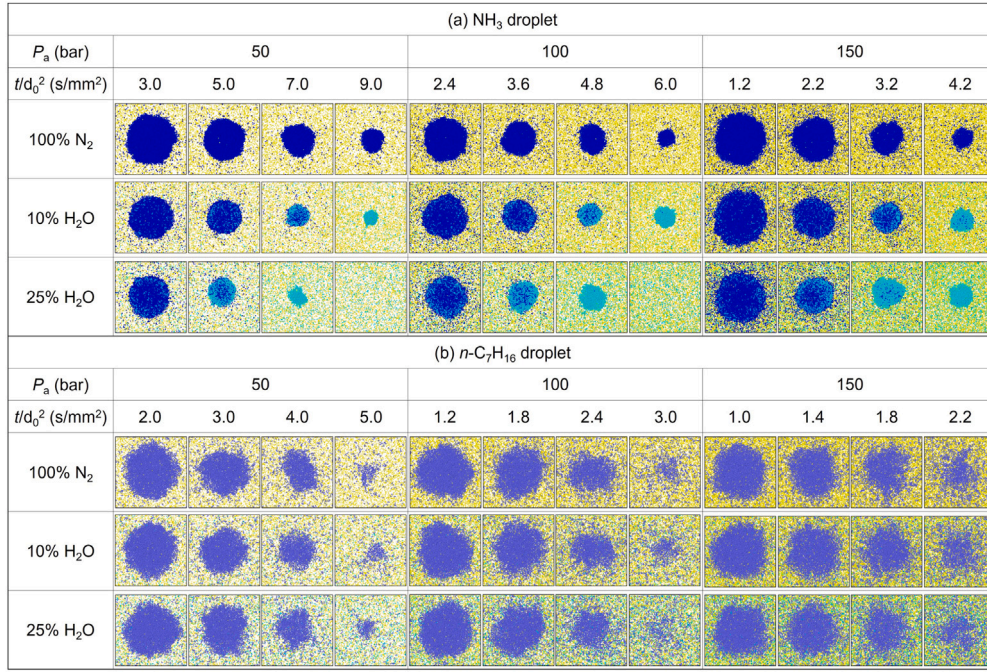


Fig. 3. Droplet cross-sectional views at $T_a = 1000$ K under varying P_a conditions and gas compositions for (a) ammonia droplets and (b) *n*-heptane droplets (ammonia molecules are depicted in dark blue, *n*-heptane molecules in purple, nitrogen molecules in yellow, and water molecules in light blue; 10% H_2O represents the gas environment of 90% N_2 +10% H_2O , and similar notation applies to other cases).

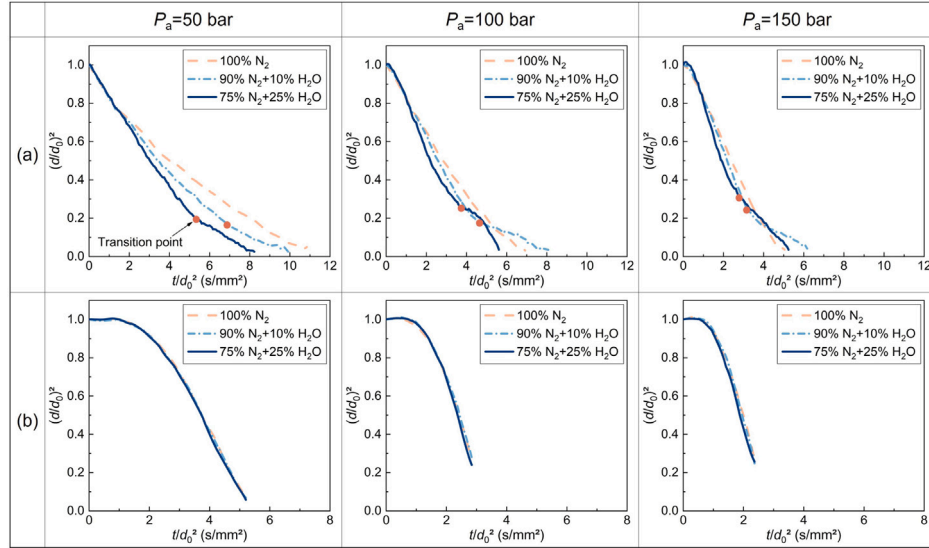


Fig. 4. Evolutions of the squared normalized droplet diameter $(d/d_0)^2$ at $T_a = 1000$ K under varying P_a conditions and gas compositions for (a) ammonia droplets and (b) *n*-heptane droplets.

This transition separates the entire evaporation process into two distinct stages, which are reflected in the $(d/d_0)^2$ curve shown in Fig. 7. The separation moment is marked as the transition point (t_T). Based on the evolution of composition fraction, the earlier stage, characterized by a higher ammonia concentration, can be defined as the ammonia-dominated evaporation (Stage 1). Once the x_{H_2O} exceeds a critical threshold (i.e., the formation of the condensation nuclei), the phase change process transitions to the water-dominated evaporation (Stage 2). It is worth noting that the slope of the $(d/d_0)^2$ curve in Stage 2 is noticeably smaller than that in Stage 1, indicating a reduced evaporation rate during the water-dominated evaporation. This difference arises from the higher boiling point and greater latent heat of vaporization of water compared to ammonia. Similar two-stage evaporation behavior

has been reported in studies of polar droplets, including ammonia [13, 14] and alcohols [15], in humid air, and the mechanisms identified in those works are in good agreement with this work. Furthermore, although water vapor accelerates ammonia evaporation in Stage 1 but inhibits it in Stage 2, the net effect is still that ammonia evaporates faster in environments containing water vapor [14].

The effects of ambient water vapor concentration and pressure on t_T are revealed in Fig. 8. It is observed that the t_T decreases with increasing water vapor concentration and pressure, suggesting that both factors can enhance the solubility of water vapor in the ammonia droplet. This increased solubility promotes earlier condensation nucleation, thereby accelerating the transition to Stage 2. These findings underscore the significant role of water vapor in influencing the phase

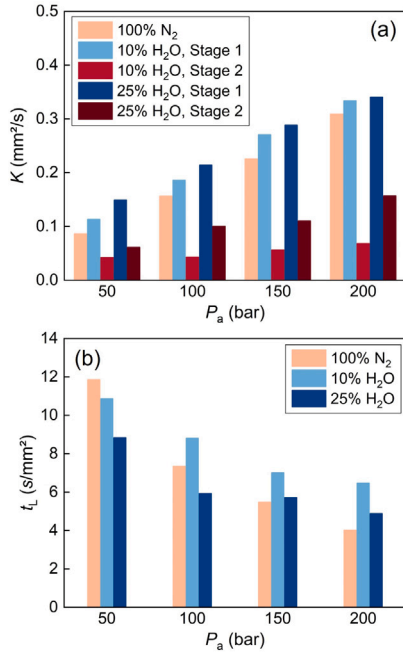


Fig. 5. Evaporation rate (K) (a) and droplet lifetime (t_L) (b) at $T_a = 1000$ K under varying P_a conditions and gas compositions for ammonia droplets.

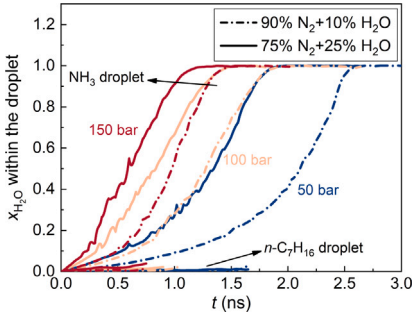


Fig. 6. Evolution of the water mole fraction (x_{H_2O}) within different droplets at $T_a = 1000$ K under varying P_a conditions and gas compositions.

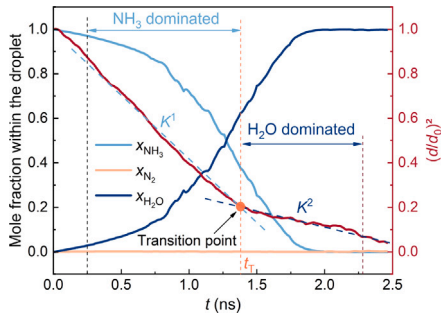


Fig. 7. Evolution of the mole fraction of each component within the ammonia droplet and the $(d/d_0)^2$ curve at $T_a = 1000$ K and $P_a = 100$ bar in the 90% $N_2 + 10\%$ H_2O environment.

change process of ammonia droplets. Consequently, to improve the predictive accuracy of liquid ammonia spray simulations, particularly under high water vapor concentration conditions, it is essential to develop a comprehensive phase change model for ammonia droplets that accounts for the dissolution and condensation effects of polar water vapor.

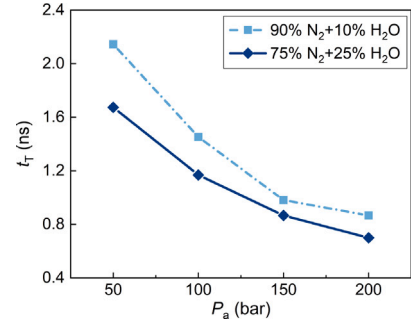


Fig. 8. Transition point moment (t_T) at $T_a = 1000$ K under varying P_a conditions and water vapor concentrations.

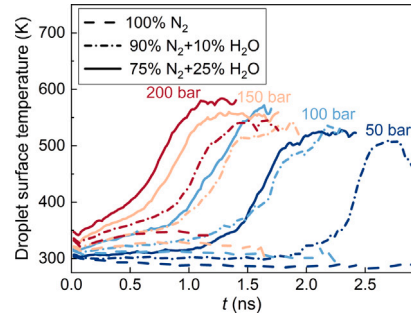


Fig. 9. Evolution of the droplet surface temperature at $T_a = 1000$ K under varying P_a and gas compositions.

3.3. Underlying mechanisms of water vapor

The underlying mechanisms by which ambient water vapor influences ammonia droplet evaporation can be further revealed from the following key perspectives. As depicted in Fig. 9, droplets in nitrogen/water environments exhibit significantly higher surface temperatures compared to those in pure nitrogen environments, with both the rate and magnitude of temperature rise increasing as the water vapor concentration increases. This outcome is primarily attributed to two factors: the release of latent heat during the dissolution and condensation of water vapor [13–15], and the enhanced heat transfer due to the inherently higher thermal conductivity of water compared to ammonia [16].

Moreover, the complex hydrogen bonding interactions of water and ammonia molecules also play a key role in the phase change process. Fig. 10 illustrates the evolution of the number and density of hydrogen bonds (HBs) within droplets under varying water vapor concentrations. In the pure nitrogen environment, the ammonia–ammonia HBs number gradually decreases as evaporation proceeds, whereas their HBs density remains relatively stable throughout most of the droplet lifetime. By contrast, as the water vapor concentration increases, the ammonia–ammonia HBs number declines more rapidly, and their HBs density also begins to decrease. Since hydrogen bonding is a key contributor to intermolecular forces, this reduction in HBs density indicates a weakening of the intermolecular forces of ammonia. In other words, the dissolution of water vapor disrupts the hydrogen bond network of ammonia, weakens the intermolecular forces, and consequently lowers the evaporation energy barrier [17]. Although the dissolution process also promotes the formation of water–ammonia HBs, this increase is much smaller than the reduction in ammonia–ammonia HBs (see Fig. 10a). Furthermore, the comparison of the HBs number evolution with the $(d/d_0)^2$ curve shown in Fig. 11 reveals that most water–ammonia HBs formation occurs in Stage 1 before the transition point, which has little effect on the evaporation rate. This is because the effect

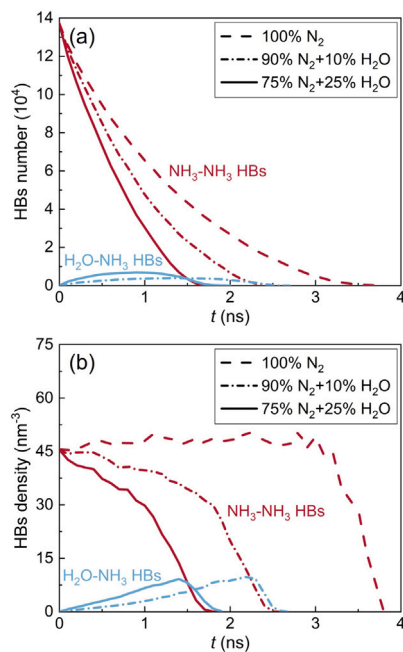


Fig. 10. Evolution of (a) HBs number and (b) HBs density of droplets at $T_a = 1000$ K and $P_a = 50$ bar under varying water vapor concentrations.

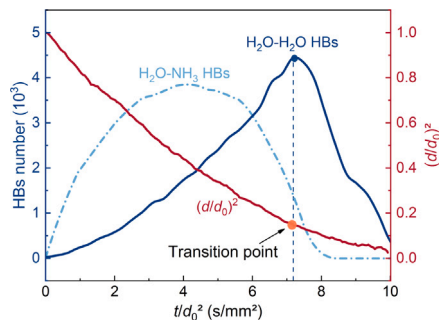


Fig. 11. Evolution of the HBs number and $(d/d_0)^2$ at $T_a = 1000$ K and $P_a = 50$ bar in the 90% $N_2 + 10\%$ H_2O environment.

caused by the water-ammonia HBs formation is much weaker than the heat transfer enhancement effect induced by water vapor dissolution. Therefore, this minor adverse effect can be neglected. In addition, Fig. 11 shows that during Stage 1, as the water molecules within the droplet condense, the water-water HBs number gradually increases. Notably, the peak moment in water-water HBs number is highly consistent with the transition point moment (t_T). This correlation indicates that the transition of the mixed droplet from Stage 1 to Stage 2 is mainly driven by the condensation nucleation behavior of water molecules. Accordingly, this water-water HBs peak point can serve as a key criterion for identifying the transition of ammonia droplet evaporation modes in environments containing water vapor. Overall, both enhanced heat transfer and weakened intermolecular forces due to the water vapor dissolution contribute to the accelerated ammonia droplet evaporation in water-rich environments.

4. Conclusions

In this study, MD simulations were conducted to comprehensively investigate the effect of water vapor on the phase change behaviors of ammonia droplets under varying ambient pressures. To improve the prediction accuracy and practicality, a new flexible potential model for

ammonia was developed based on first-principles calculations. Using this model, the unique phase change behavior of ammonia in the nitrogen/water environment was analyzed. Additionally, the underlying mechanisms by which water vapor influences droplet evaporation were also elucidated. The main conclusions can be drawn as follows:

- A flexible potential model for ammonia was developed based on first-principles calculations. This model demonstrates high accuracy in predicting the thermodynamic and transport properties of ammonia, as well as multi-component VLE predictions.
- Both elevated ambient pressures and increased water vapor concentrations can promote the ammonia droplet evaporation. In nitrogen/water environments, ammonia droplets exhibit unique phase change behaviors that differ notably from those of non-polar n -alkane droplets. During evaporation, polar water vapor dissolves into the polar ammonia droplet and subsequently condenses, eventually nucleating into a small water droplet.
- The dissolution and condensation of water vapor divide the ammonia droplet evaporation process into two stages: an ammonia-dominated evaporation, followed by a water-dominated evaporation. Increased initial water vapor concentration and higher ambient pressure enhance water solubility within the ammonia droplet, thereby accelerating both ammonia evaporation and water condensation.
- The mechanism by which water vapor promotes the ammonia droplet evaporation is mainly because, during dissolution and condensation, water releases latent heat and increases droplet thermal conductivity, raising droplet temperature. Meanwhile, water weakens the hydrogen bonding among ammonia molecules, lowering the evaporation energy barrier.

Novelty and significance statement

The novelty of this study lies in the detailed investigation of the unique phase change behavior of ammonia in environments containing water vapor at the atomic level through molecular dynamics simulations. Furthermore, the mechanisms by which water vapor affects the ammonia droplet evaporation are comprehensively evaluated.

It is significant because ammonia is a highly promising zero-carbon fuel, and its clean and efficient utilization is crucial for energy systems. Nevertheless, the phase change behavior of ammonia significantly differs from that of traditional hydrocarbon fuels. Notably, a late injection strategy for liquid ammonia is usually adopted to improve combustion stability in ammonia-fueled engines. Upon this, the phase change process of ammonia will occur in the presence of water vapor, a major combustion product. However, phase change behaviors of ammonia droplets in gas environments containing water under engine-relevant conditions still lack systematic research, and the relevant understanding is still insufficient. This research serves as a useful reference for predicting the fuel/air mixing characteristics of liquid ammonia under complex gas environments under engine-relevant conditions.

CRediT authorship contribution statement

Yanzhi Zhang: Writing – review & editing, Writing – original draft, Software. **Feilong Chen:** Methodology, Software, Investigation, Validation, Writing – original draft, Writing –review & editing. **Xuehao Zhang:** Methodology, Software, Validation. **Ming Jia:** Project administration, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported by the National Natural Science Foundation of China (Grant No. 52106154 and 52325603) and Open Project of the State Key Laboratory of Engines (Tianjin University) (Grant No. K2025-06).

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.proci.2025.105897>.

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